AI Agent Project Guide

16 portfolio use cases · difficulty-wise · Assistant / Coworker / Ambient · local models and free API options

Your machine	Mac M3 · 18 GB unified memory	**Suggested language**	Python
Best local runtime	Ollama	**Best first project**	#16 Product Expert & Quoting

How to read the agent types

TYPE	MEANING	TYPICAL TRIGGER	EXAMPLE FROM YOUR SLIDES
Assistant	Responds when a person asks; supports decisions.	User prompt / chat request	#6 Forecast analyst copilot; #11 Service-lane copilot
Coworker	Completes a bounded multi-step task using tools and data; human reviews important outputs.	Assigned task / uploaded file / request	#3 Loan-file prep; #7 Due diligence; #9 Governance auditor; #10 Negotiation prep
Ambient	Runs in the background, watches events/data, and proactively routes or acts within guardrails.	New event / schedule / threshold	#2 Patient triage; #4 Claims triage; #5 Schedule watcher; #8 L1 IT support; #12 Guest agent; #13 Ad watcher
Reserve	Reserve label from the source slides, not an autonomy category. Treat it as a backlog / later candidate.	Depends on implementation	#14 Admissions; #15 Caseworker; #16 Product expert

Important: Difficulty below means a small learning/prototype build using sample data. A production healthcare, lending, insurance, or government agent is much harder because of privacy, security, auditability, integrations, and human oversight.

All 16 projects — sorted by prototype difficulty

The category labels reproduce the slides. Reserve is shown separately because it is a portfolio status, not an agent operating mode.

Easy projects

#	PROJECT / SECTOR	WHAT IT DOES	TYPE	PYTHON BUILD IDEA
11	**Service-lane copilot** Automotive · Dealerships	Uses vehicle service history to suggest relevant next services for an advisor to approve.	**Assistant**	Search sample service history + maintenance rules; explain recommendations with source records.
14	**Admissions counsellor** Education · Admissions	Answers applicant questions from programme rules and tracks each applicant's progress.	**Reserve**	University-policy PDF RAG + FAQ chat; add application tracker later.
16	**Product expert & quoting** Manufacturing · B2B sales	Answers technical product questions from manuals and drafts quotes for sales approval.	**Reserve**	Product-manual RAG + catalogue/price lookup tool + quote calculator/template; approval before sending.

Medium projects

#	PROJECT / SECTOR	WHAT IT DOES	TYPE	PYTHON BUILD IDEA
3	**Loan-file prep agent** Financial services · Lending	Extracts information from loan documents, checks required items and policy rules, and drafts an underwriter memo.	**Coworker**	PDF/OCR extraction + structured JSON + deterministic validations + cited assessment.
5	**Schedule-risk watcher** Construction · Scheduling	Monitors schedule and field updates, predicts delays, and suggests resequencing for a project manager to approve.	**Ambient**	Read project CSV/API updates on a schedule; calculate slippage; use LLM to explain risks and draft options.
6	**Forecast analyst copilot** Energy · Power markets	Explains load and price forecasts from an existing model and drafts a daily trading-desk brief.	**Assistant**	Load sample time-series/forecast output; Python/Pandas analysis; LLM narrates findings and caveats.
7	**Due-diligence coworker** Real estate · CRE intelligence	Compiles leases, comparable properties, and risk flags into a cited investment memo.	**Coworker**	Multi-PDF ingestion + RAG + structured comparison table + cited memo; use trusted data.
8	**L1 IT support agent** Enterprise IT · Help desk	Triages tickets, resolves low-risk requests, and escalates other issues with diagnostics attached.	**Ambient**	Ticket trigger + knowledge-base RAG + allow-listed tools; require approval for account/security changes.
10	**Negotiation prep coworker** Business ops · Procurement	Compares supplier quotes and purchasing history and drafts a negotiation brief with target prices.	**Coworker**	Parse quote spreadsheets/PDFs; normalize units; calculate comparisons in Python; draft negotiation brief.
12	**Guest message & offer agent** Hospitality · Direct booking	Answers guest enquiries and proposes personalized offers; discounts need approval.	**Ambient**	Incoming-message trigger + property FAQ RAG + offer rules; draft replies and request approval for discounts.
13	**Ad performance watcher** Media · Ad creative ops	Flags CPA/ROAS anomalies in daily campaign data and proposes creative changes for approval.	**Ambient**	CSV ingestion + Pandas anomaly thresholds + LLM summary and suggested experiments.

Hard projects

#	PROJECT / SECTOR	WHAT IT DOES	TYPE	PYTHON BUILD IDEA
1	**Care-gap closer** Healthcare · HEDIS quality	Finds missing quality-measure activities in patient records and prepares outreach/evidence for clinician review.	**Coworker**	RAG over approved clinical records; rules engine for measure checks; draft outreach; clinician approval.
2	**Patient-message triage** Healthcare · Telehealth	Sorts patient portal messages by urgency, routes them, and drafts replies; it must not diagnose.	**Ambient**	Message trigger + classifier + routing rules + safe draft response; urgent cases go to clinical staff.
4	**Claims triage agent** Insurance · Claims intake	Classifies new claims, flags fraud indicators, requests missing documents, and leaves the decision to an adjuster.	**Ambient**	Intake trigger + extraction + rules/ML risk indicators + human review; never auto-deny claims.
9	**Agent governance auditor** Enterprise IT · AI governance	Reviews agent traces and MCP configurations against security policies and reports findings.	**Coworker**	Ingest synthetic traces/configs; evaluate against a checklist; produce evidence-linked findings.
15	**Caseworker coworker** Government · Reentry services	Matches clients to housing and job options and drafts a case plan for a caseworker.	**Reserve**	Search curated resource directory + eligibility rules + draft case plan; human review and privacy controls.

Reserve projects & how to choose

#	PROJECT / SECTOR	DESCRIPTION	TYPE	SIMPLE PROTOTYPE APPROACH

Suggested shortlist for your Mac M3 (18 GB)

PROJECT	WHY PICK IT
#16 Product Expert & Quoting	Best all-round starter: PDFs + RAG + product lookup + quote calculation.
#14 Admissions Counsellor	Simplest RAG project: answer from a small set of policy/brochure PDFs with citations.
#11 Service-lane Copilot	Good structured-data project: service history + deterministic maintenance rules.
#10 Negotiation Prep	Good next step: extract quote data from PDFs/spreadsheets and compare with Python.
#8 L1 IT Support	Best tool-use demonstration: search KB, create/read tickets, and escalate.

Learning path: start as an Assistant (chat + RAG), add Python tools and approvals to become a Coworker, then add event triggers, scheduling, monitoring, and guarded actions to become Ambient.

Local models for your Mac M3 · 18 GB unified memory

These models run on your Mac after downloading. They do not require an inference API key and can work offline once model files and dependencies are installed.

MODEL FAMILY	TRY THIS SIZE	GOOD FOR	NOTES FOR YOUR MAC
Qwen 3	4B first; 8B if responsive	General agent tasks, structured extraction, tool-use experiments	Recommended starting family. 8B quantized is a reasonable balance; monitor memory and context length.
Gemma 3	4B; test 12B cautiously	General assistant; vision-capable variants for image/document experiments	4B is safer with 18 GB. 12B may leave less memory for macOS and other services.
Llama 3.2	3B	Lightweight chat, classification, routing prototypes	Good fallback when speed and memory matter more than complex reasoning.
Mistral 7B	7B quantized	General text tasks and document Q&A;	A useful alternative to compare against Qwen; quality varies by task.
DeepSeek R1	7B / 8B distilled variant	Reasoning-heavy analysis and step-by-step task prototypes	Can be slower and use more tokens. Keep context moderate.

Useful local embedding models

nomic-embed-text	Local text embeddings for semantic search / RAG.
all-MiniLM-L6-v2	Small Sentence Transformers embedding model; convenient for lightweight prototypes.
BAAI/bge-small-en-v1.5	Compact English embedding model for retrieval experiments.

Ollama commands to get started

```
# Install Ollama for macOS: https://ollama.com/download/mac
ollama pull qwen3:4b
ollama pull nomic-embed-text
ollama run qwen3:4b
```

Practical memory note: model file size is not the same as total runtime memory. macOS, context/KV cache, embeddings, ChromaDB, Python, and your UI also consume unified memory. Start with 4B, then benchmark 8B with a moderate context window. Avoid trying to keep several large models loaded simultaneously.

Hosted APIs with free or limited-free access

These run in the provider's cloud. They are not local/offline, and free plans have quotas, model availability restrictions, and changing terms. Use them for prototypes; never put API keys in frontend code.

PROVIDER	FREE ACCESS TYPE	GOOD FOR	CAVEATS	OFFICIAL LINK
Google Gemini API	Ongoing free tier for selected models	Strong general assistant, document Q&A;, multimodal prototypes	Model-specific RPM/TPM/RPD limits; free-tier data-use terms differ from paid tier.	Pricing / free tier
GroqCloud	Free developer tier with per-model limits	Fast inference, tool-use prototypes, classification	Rate limits vary by model and account; check console limits. Free tier is not production SLA.	Rate limits
Mistral API	Free experimentation mode / limited access	Text generation, extraction, classification	Free-mode model access and quotas are restricted; verify current account limits.	Docs
OpenRouter	Selected model IDs ending in :free	Trying different hosted models through one API	Free models are rate-limited; published limits include a daily cap. Model availability changes.	Free models
Cloudflare Workers AI	10,000 Neurons/day free allocation	Hosted inference and experimentation across catalog models	Quota is measured in Neurons, not requests; some models require paid plan.	Pricing

What about Grok / xAI?

Grok is available as a consumer chatbot with limited free access, but the xAI developer API is priced by token/model and should be treated as paid API usage—not as a dependable free API tier. Check the current model and pricing pages before using it.

Official references: xAI API pricing · xAI models

Other options to investigate

Cohere: trial/developer access may be available, particularly useful for embeddings and reranking; verify whether the current key and intended use are eligible. **NVIDIA NIM:** developer catalog access may include limited trial/prototyping quotas for selected models. These should be treated as trial/limited access, not unlimited free production APIs.

Avoid relying on stale API lists: free tiers and model IDs change frequently. Confirm the provider's official pricing, rate limits, data-use terms, and allowed commercial use immediately before implementation.

Python stack & implementation roadmap

LAYER	RECOMMENDED PYTHON TOOL	WHAT IT DOES
Local model runtime	Ollama + ollama-python	Runs local chat and embedding models; local HTTP endpoint for your Python app.
Backend API	FastAPI + Pydantic	Expose chat, upload, search, tool, and workflow endpoints.
PDF / files	PyMuPDF; python-docx; pandas	Extract text and tabular data from source files.
RAG / vector search	ChromaDB + local embeddings	Chunk, embed, retrieve relevant source passages, and provide citations.
Agent workflows	Start with Python functions; add LangGraph	Control steps, branching, retries, state, and human approval.
Frontend	Streamlit first; React later	Quick prototype UI; React if you want a polished product interface.
Persistence	SQLite first; PostgreSQL later	Store products, users, chat sessions, generated quotes, and audit events.

Recommended build sequence for #16 Product Expert & Quoting

1 · Assistant	Upload product manuals; extract/chunk text; retrieve passages; answer questions with page/source references.
2 · Coworker	Add Python tools: search catalogue, check stock, calculate totals/discounts, create a quote draft. Require approval before sending.
3 · Ambient	Monitor an inbox or request queue; detect new quote requests; prepare quote drafts automatically; notify a salesperson for approval.

Safety and quality guardrails

For healthcare, lending, insurance, employment/housing, and other high-impact use cases: use synthetic or properly authorized data for a prototype; keep a human responsible for consequential decisions; enforce access control; log tool calls; validate calculations in deterministic Python code; and never let an LLM invent policy, pricing, medical advice, or eligibility decisions.

Reference links

Ollama model library: ollama.com/library · Gemini API pricing: Google AI pricing · Groq limits: Grog rate limits · OpenRouter free models: OpenRouter · Cloudflare AI pricing: Cloudflare · xAI pricing: xAI

Prepared 2 November 2026. Model names, free-tier access, quotas, and terms may change; re-check provider documentation before deployment.